

Gold Store ERP – Professional ERD

Identity Module

Users

- Id (PK)
 - DisplayName
 - Email
 - PhoneNumber
 - PasswordHash
 - IsActive
 - CreatedAt
-

Roles

- Id (PK)
 - Name
-

Permissions

- Id (PK)
 - Name
-

UserRoles

- UserId (FK → Users)
- RoleId (FK → Roles)

Composite PK:

(UserId, RoleId)

RolePermissions

- RoleId (FK → Roles)
- PermissionId (FK → Permissions)

Composite PK:

(RoleId, PermissionId)

Configuration Module

Categories

(Self-reference hierarchy)

- Id (PK)
 - ParentCategoryId (FK → Categories)
 - Name
 - SortOrder
-

Karats

- Id (PK)
- Value

Examples:

18

21

24

GoldPrices

- Id (PK)
- PriceDate
- Price21
- Currency
- CreatedByUserId (FK → Users)

Unique:

(PriceDate, Currency)

Finance Module

Accounts

Represents:

- Cash Accounts
 - Bank Accounts
 - Id (PK)
 - Name
 - AccountType
 - Currency
 - BankName
 - AccountNumber
 - IsActive
-

FinancialLedgerEntries

Unified Money Ledger

- Id (PK)
- AccountId (FK → Accounts)
- Currency
- Amount
- TransactionDirection

(IN / OUT)

- TransactionType
- ReferenceType
- ReferenceId
- Notes
- CreatedByUserId
- CreatedAt

Indexes:

(AccountId)

(TransactionType)

(ReferenceType, ReferenceId)

(CreatedAt)

Supplier Module

Suppliers

- Id (PK)
 - Name
 - PrimaryPhone
 - SecondaryPhone
 - BankAccountNumber
 - Notes
-

SupplierDeliveries

(Header)

- Id (PK)
 - SupplierId (FK)
 - DeliveryDate
 - FeePerGram
 - TotalManufacturingFee
 - ReceivedByUserId
 - Notes
-

SupplierDeliveryItems

- Id (PK)
 - SupplierDeliveryId (FK)
 - Weight
 - KaratId
 - Equivalent21Weight
-

SupplierPayments

(Header)

- Id (PK)
- SupplierId (FK)
- PaymentDate

- Notes
 - ReceivedByUserId
-

SupplierScrapPaymentItems

- Id (PK)
 - SupplierPaymentId (FK)
 - Weight
 - KaratId
 - Equivalent21Weight
-

SupplierManufacturingPayments

- Id (PK)
 - SupplierPaymentId (FK)
 - AccountId (FK)
 - Currency
 - Amount
-

Sales Module

SalesInvoices

(Header)

- Id (PK)
- InvoiceNumber
- InvoiceDate
- CustomerName
- CustomerBankAccount
- SellerUserId
- Notes

Unique:

InvoiceNumber

SalesInvoiceItems

- Id (PK)

- SalesInvoiceId (FK)
 - CategoryId (FK)
 - Weight
 - KaratId
 - GramPrice
 - Currency
 - TotalAmount
-

SalesInvoicePayments

- Id (PK)
 - SalesInvoiceId (FK)
 - AccountId (FK)
 - PaymentMethod
 - Currency
 - Amount
-

Customer Gold Purchase Module

CustomerPurchaseInvoices

(Header)

- Id (PK)
 - InvoiceNumber
 - PurchaseDate
 - CustomerName
 - NationalId
 - Phone
 - Address
 - BirthYear
 - BankAccount
 - BuyerUserId
 - Notes
-

CustomerPurchaseItems

- Id (PK)
- CustomerPurchaseInvoiceId (FK)
- CategoryId
- Weight
- KaratId

- GramPrice
 - Currency
 - TotalPrice
-

CustomerPurchasePayments

- Id (PK)
 - CustomerPurchaseInvoiceId (FK)
 - AccountId
 - Currency
 - Amount
-

Inventory Module

GoldLedgerEntries

Unified Gold Ledger

- Id (PK)
- KaratId (FK)
- Weight
- Equivalent21Weight
- MovementDirection

(IN / OUT)

- MovementType

Examples:

SupplierDelivery

CustomerPurchase

Sale

SupplierScrapPayment

InventoryAdjustment

- ReferenceType
- ReferenceId
- CreatedByUserId

- CreatedAt

Indexes:

(KaratId)

(MovementType)

(ReferenceType, ReferenceId)

(CreatedAt)

InventoryAdjustments

(Header)

- Id (PK)
 - AdjustmentDate
 - AdjustmentReason
 - Notes
 - CreatedByUserId
-

InventoryAdjustmentItems

- Id (PK)
- InventoryAdjustmentId
- KaratId
- Weight
- Equivalent21Weight
- AdjustmentType

Increase

Decrease

Damage

Loss

Correction

Expense Module

ExpenseCategories

- Id (PK)
 - Name
-

Expenses

- Id (PK)
 - ExpenseCategoryId
 - AccountId
 - Currency
 - Amount
 - Description
 - ExpenseDate
 - CreatedByUserId
-

HR Module

Employees

- Id (PK)
 - Name
 - Phone
 - Position
 - Salary
-

SalaryPayments

- Id (PK)
 - EmployeeId
 - AccountId
 - Currency
 - Amount
 - SalaryMonth
 - PaymentDate
-

Suggested Views

CurrentGoldStockView

Grouped By:

- Karat

Returns:

- CurrentWeight
 - Equivalent21Weight
-

AccountBalanceView

Grouped By:

- Account

Returns:

- CurrentBalance
-

SupplierGoldBalanceView

Returns:

Supplier Delivered Gold

minus

Supplier Scrap Payments

SupplierManufacturingBalanceView

Returns:

Manufacturing Fees

minus

Main Relationships

Supplier 1 → Many SupplierDeliveries

SupplierDelivery 1 → Many SupplierDeliveryItems

Supplier 1 → Many SupplierPayments

SalesInvoice 1 → Many SalesInvoiceItems

SalesInvoice 1 → Many SalesInvoicePayments

CustomerPurchaseInvoice 1 → Many CustomerPurchaseItems

CustomerPurchaseInvoice 1 → Many CustomerPurchasePayments

Account 1 → Many FinancialLedgerEntries

Karat 1 → Many GoldLedgerEntries

Employee 1 → Many SalaryPayments

Category (Self-reference)

User 1 → Many Transactions